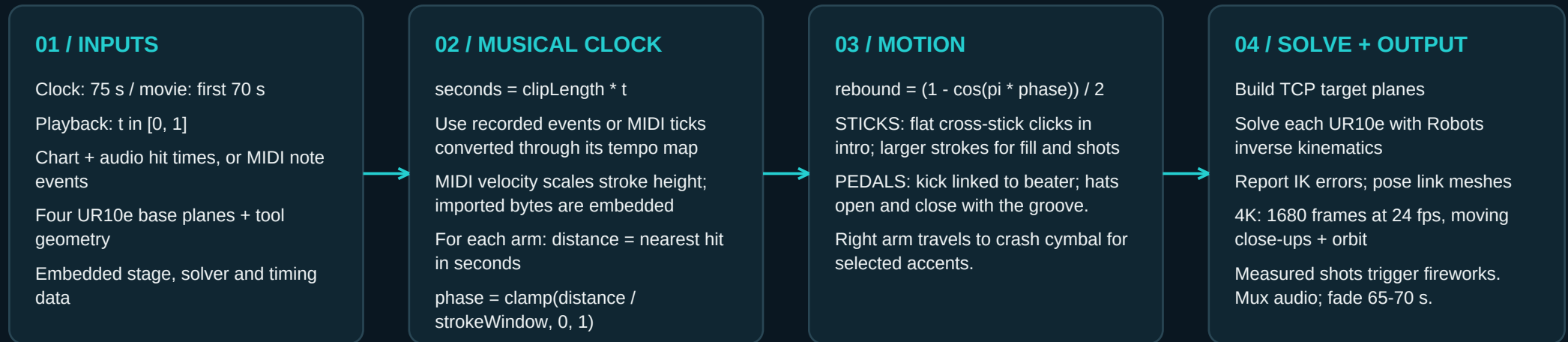


BIRDLAND / FOUR ROBOT ARMS

KINEMATIC DRUM STUDY / RHINO 8 + GRASSHOPPER + ROBOTS 2.4.0



LOOP: update playback -> recompute clock -> swing / press -> solve -> display

DESIGN INTENT

Four robot arms reinterpret the hand and foot roles of a drummer in the opening of Weather Report's Birdland: hi-hat strokes, flat snare side-stick clicks, a linked kick beater and a hi-hat pedal. Chart-guided audio events drive the internally referenced Grasshopper motion, while the reusable MIDI variant embeds imported note data and maps tempo and velocity to the same arms. The 4K film alternates an orbit with instrument close-ups; measured ensemble attacks trigger fireworks. These presentation effects emphasize the musical structure, with original audio and a fade from 1:05 to 1:10.

VALIDATION + BOUNDARIES

Birdland motion has been sampled across 1800 instants; the MIDI demo across 217. This is an inverse-kinematics visualization, not a controller-ready program. Continuous collisions, joint speeds, accelerations, contact forces and physical synchronization require validation before execution.

DELIVERABLES: MP4 with original audio / Internally referenced GH / This diagram + intent

Audio: Weather Report official channel (SvhmaNlGgRM), 00:00-01:10. Chart: Pedro Marambio / Drumeo.